

Subauroral plasma boundaries and associated magnetospheric phenomena

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Introduction

The plasmasphere — a torus of cold corotating plasma "frozen" into Earth's magnetic field and sourced from the ionosphere — is a key subsystem of the complex geospace environment. Its outer boundary, the plasmopause, is marked by an abrupt density drop. During geospace storms, enhanced convection removes the outer shells of the plasmasphere, and drainage channels called plasmaspheric plumes form, transporting plasmaspheric plasma toward the dayside magnetopause. Ionospheric signatures of the plasmopause and the plumes have been identified in the main ionospheric trough (MIT) and the post-noon storm-enhanced density (SED) plumes, respectively. Recent studies have shown that the plasmopause–MIT conjugacy does not hold in general but is limited to the night- and dawn-side MLT sectors. Moreover, the plasmaspheric plume–SED plume conjugacy has not been found to hold under any conditions. On the dusk side, a signature is likewise found in the MIT, where it separates from the ionospheric footprint of the plasmopause. The subauroral electric field is the common driver underlying the formation of all these plasma structures: the plasma drift it drives supports the development of both the MIT and the SED plumes in the ionosphere, the formation of the narrow plasmaspheric plumes, and the sharpening of the newly formed plasmopause in the magnetosphere. Based on automated Swarm MIT detections, supplemented by other boundary measurements, we have established a proxy for the midnight plasmopause location, which is provided for the full mission lifetime at the ESA Space Weather Service Network Portal.

In-situ observations in the magnetosphere and the ionosphere

To investigate the spatio-temporal relationship between the plasmopause, plasmaspheric plumes and the main ionospheric trough (MIT), utilising observations from

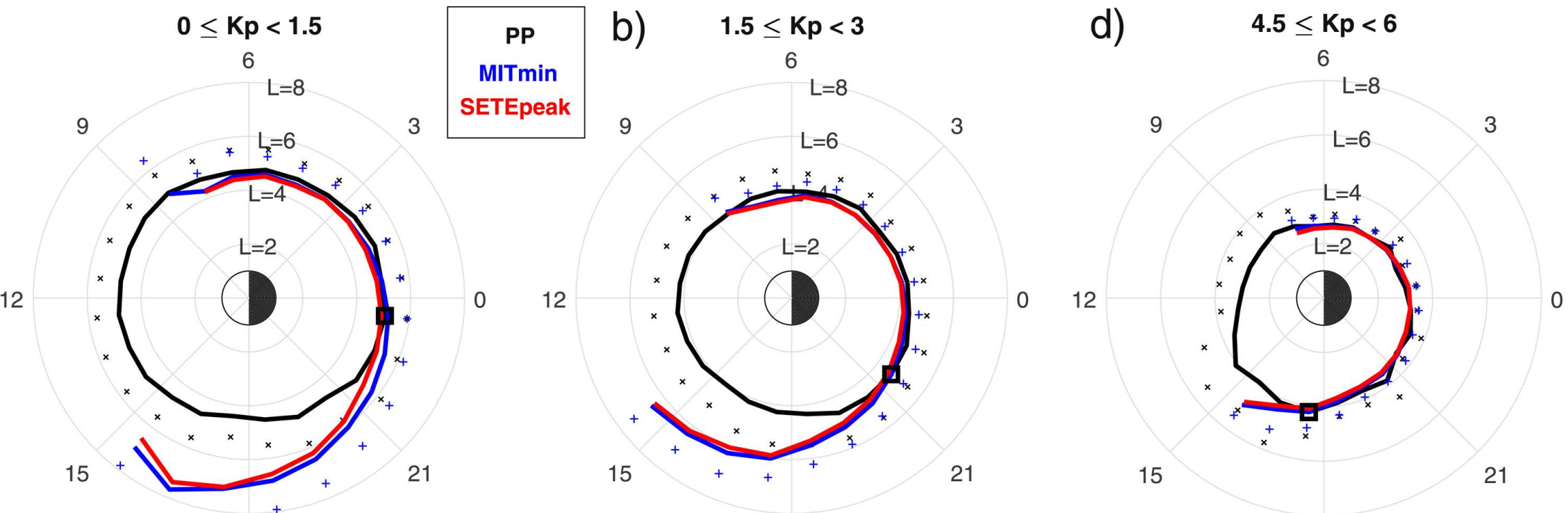
- NASA's Van Allen Probes (RBSP): plume list of Hartley et al. (2022): 869 plume crossings and
- RBSP, Arase and THEMIS plasmopause crossings: 10946 PP crossings
- ESA's Swarm mission: L2 MIT product (Heilig et al., 2022).

Regarding plume-MIT comparison, we started from the Hartley list of plume crossing events. All Swarm MIT observations within 140 min (~one and a half Swarm orbital periods) of each RBSP plume crossing were paired with the corresponding plume event. This approach resulted in 7,695 paired magnetospheric-ionospheric observations across all satellites, enabling a meaningful comparison between these boundaries.

Statistical boundaries

1) Plasmopause – Main Ionospheric Trough (night and dawn side)

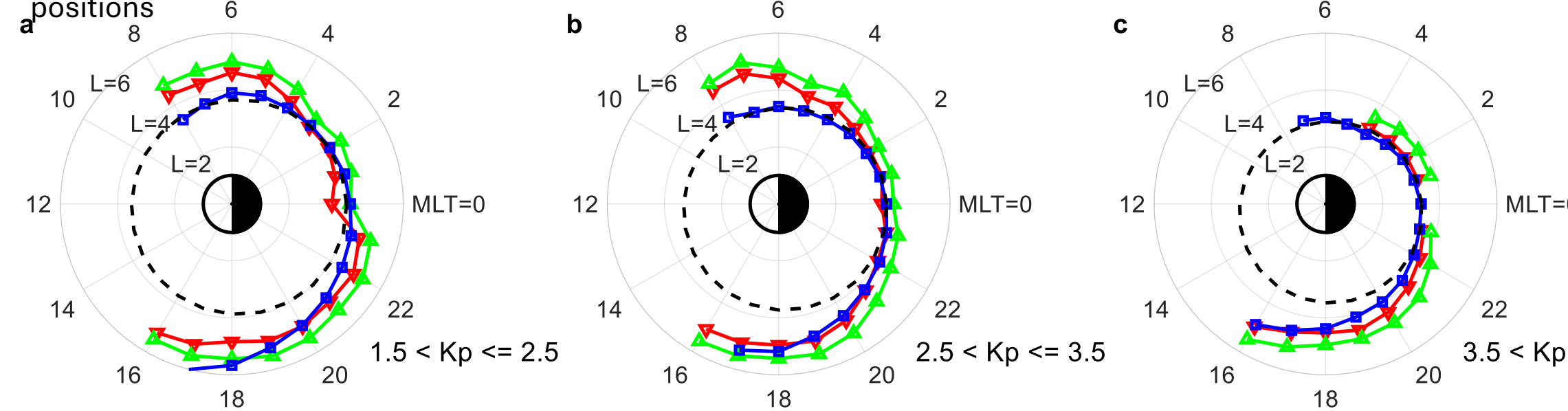
Comparison of the spatial relationship between in-situ plasmopause crossings and mapped trough (MIT) positions



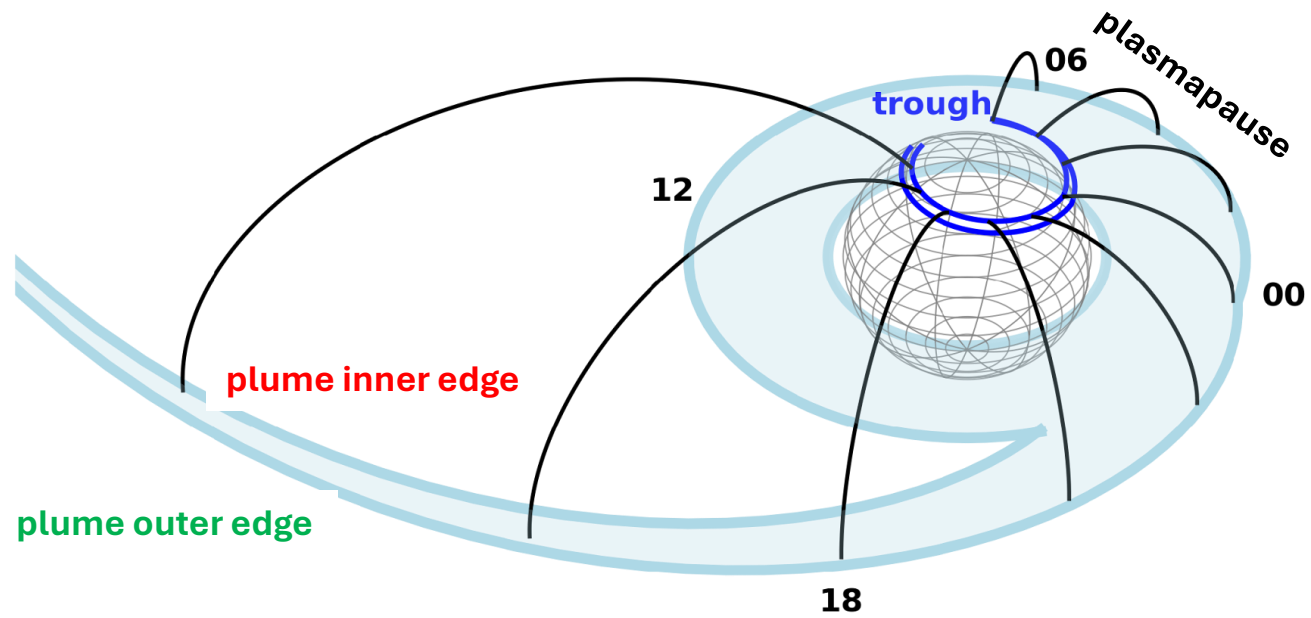
- **Plasmopause** and **mapped MIT minimum locations** are coincident (conjugate) on the night and dawn side at any level of geomagnetic activity
- mapped MIT minima separates from the plasmopause at an MLT dependent position (plasmaspheric bulge)

2) Plasmaspheric Plume – Main Ionospheric Trough (dusk side)

Comparison of the spatial relationship between in-situ plasmaspheric plume crossings and mapped trough (MIT) positions



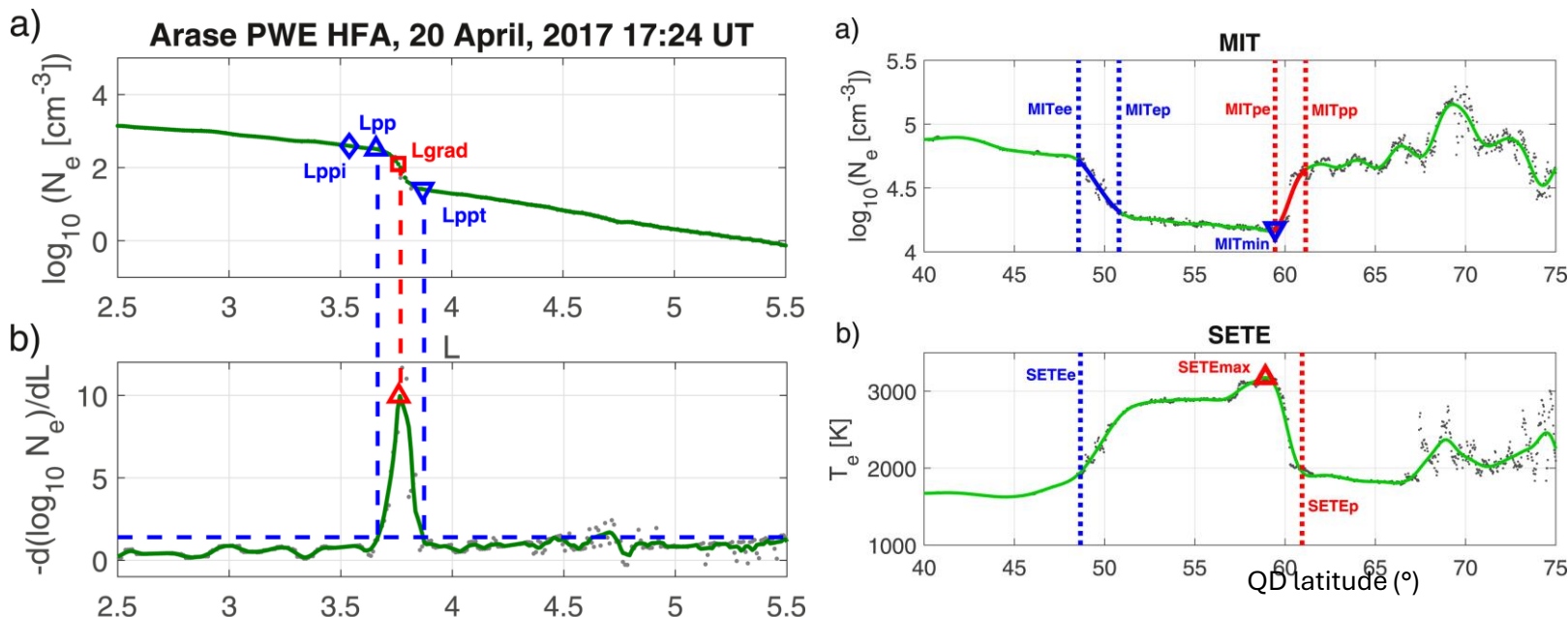
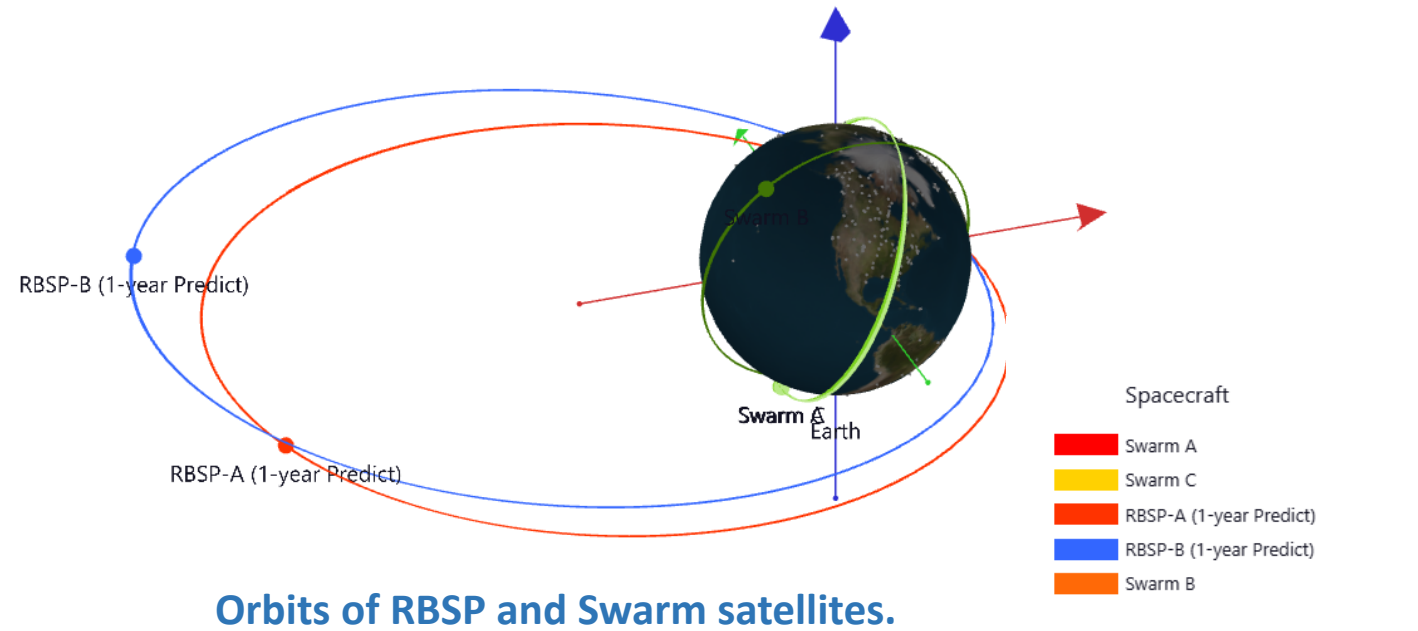
- **Plasmaspheric plume outer** and **inner edge** and **mapped MIT minimum locations** are ~conjugate on the dusk side at any level of geomagnetic activity
- SED plumes (not shown) are often observed conjugate to plasmaspheric plumes in the post-noon MLT sector



Conjugacy?

Although the same mesoscale electric field (SAPS E field) plays a role in the formation of both ionospheric (MIT, SED plume) and plasmaspheric (plume) plasma structures, the spatial relation of these structures cannot be described simply as conjugate. The reason behind this is the differential drift speed in the different regions of geospace and the variability of the driving E field. The drift speed magnitude is E/B , the angular drift speed is $E/(Br)$; however, E and Br scale differently along the field lines. Plasma parcels conjugate at some moment will not be conjugate some time later. Even in a steady flow case, the relatively faster plasmaspheric plume reaches a higher radial distance during the same time. That is why mapped MIT minima shift inward of the plume. A similar tendency must hold for the spatial relationship of plasmaspheric and SED plumes

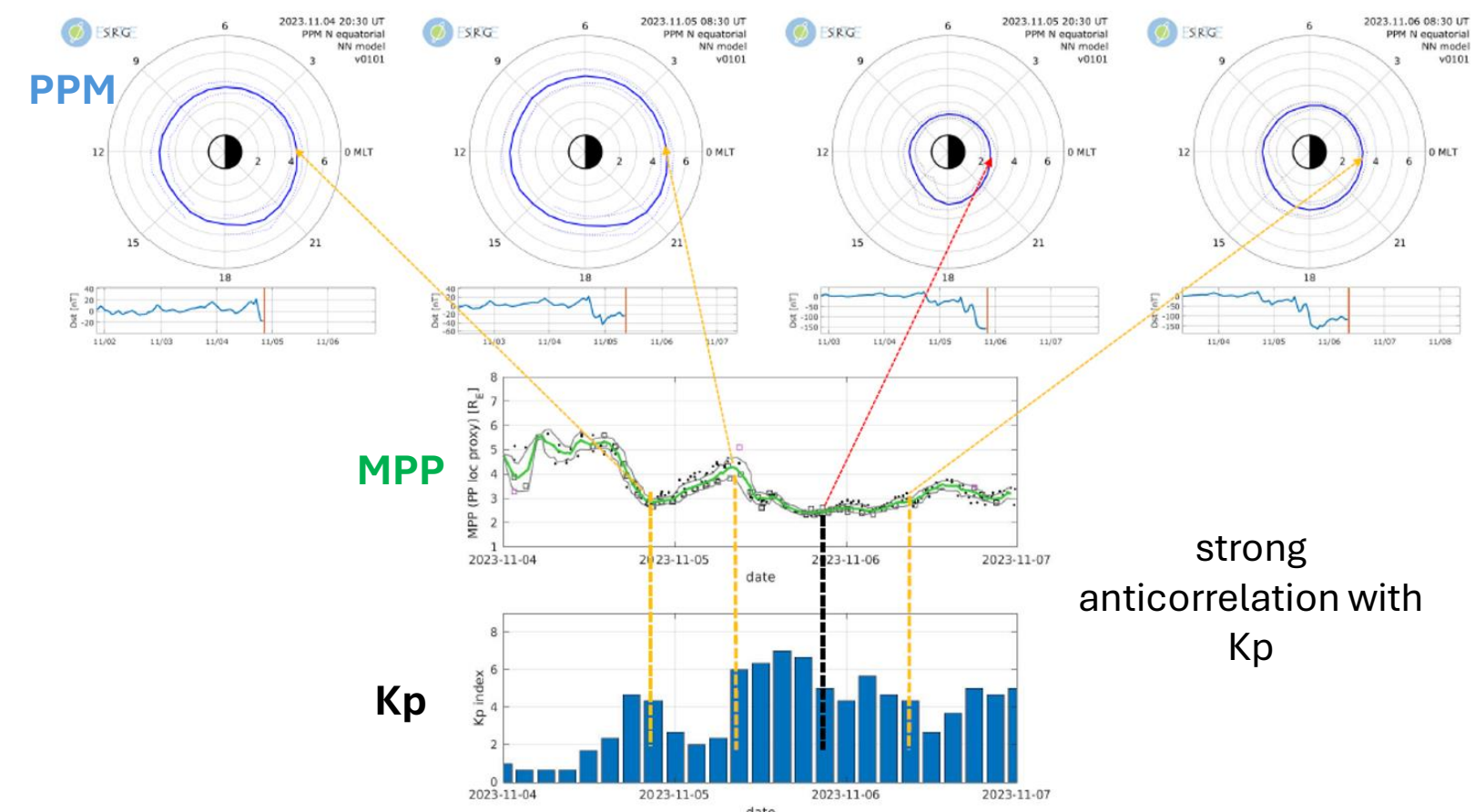
Our findings thus suggest that the concept of geospace plumes (contiguous 3D plasma structures that span along the field lines from ionosphere to ionosphere as global high-density drift shells) is incomplete and hence needs to be revised and refined.



In situ detection of plasmopause crossings in RBSP and Arase data In situ detection of the MIT structure and the associated Te enhancement in Swarm data

Midnight plasmopause location – a Swarm-based proxy

- Midnight PlasmaPause location proxy (MPP)
- Based on **MIT observations reduced to midnight MLT** and other boundary, the small-scale field-aligned current boundary (not discussed here, see Heilig and Lühr, 2013; 2018)
- Below, it is compared with a NN plasmopause model (PPM)
- Both MPP and PPM are available at ESA SWE Portal
- MPP is available from 2014 up to now, with a few-day latency



Acknowledgement

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